



IT0049

(WEB SYSTEM TECHNOLOGIES) TECHNICAL SUMMATIVE ASSESSMENT

2

TASKS FOR TODAY MANAGEMENT SYSTEM: FULL CRUD AND AUTHENTICATION

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Members (if Group):	Individual activity
Section:	TW32
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PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]
- Design program flow and application structure for computing problems using CodeIgniter and Laravel frameworks. [CLO: 3]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Extend an existing data-backed application with validated create, update, and delete workflows.
- Protect management actions behind authentication while keeping read-only pages public.
- Apply soft deletion instead of permanently removing records.

BACKGROUND INFORMATION

This assessment brings the Tasks for Today Management System from TSA1 up to the same standard as the POS application: validated forms, complete CRUD, and authentication. Permanently deleting records is generally discouraged in production systems, so this activity uses soft deletion — marking a record as archived instead of removing the row — which is the same practice discussed for the POS application in Module 3.

GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (35 points max)	All required pages, records, and workflows are present and behave as specified in the activity instructions.
Code Structure & Organization (20 points max)	The application follows the framework's MVC conventions with a clear, consistent separation of concerns across controllers, models, and views.
Validation & Security (25 points max)	Forms validate required fields and reject bad input, passwords are hashed and verified correctly, and management actions are reliably restricted to logged-in users.
Documentation & Submission Quality (20 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Extend the Tasks for Today Management System from TSA1 with full CRUD for tasks and an authentication requirement for managing them.

1. Add a password column to the users table and set a hashed password for the demo user.
2. Build a login page and a logout action. The Welcome, Task List, Profile, and About pages remain viewable by anyone, but creating, editing, or deleting a task requires being logged in.
3. Add a New Task form (/tasks/new) that validates required fields (title required, task_date required).
4. Add an edit/update page for existing tasks.
5. Add an is_archived column to the tasks table (default false). Implement a "delete" action that sets this flag instead of removing the row, and exclude archived tasks from the Welcome and Task List pages.
6. Test your access control the same way as in TFA4: confirm protected actions redirect to login when logged out, and work normally when logged in.

Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export or migration and seeder files.
- A link to your hosted, working version of the application.

REFERENCES

- CodeIgniter User Guide. Validation. <https://codeigniter4.github.io/userguide/libraries/validation.html>
- CodeIgniter User Guide. Controller Filters. <https://codeigniter4.github.io/userguide/incoming/filters.html>

SYSTEM IMPLEMENTATION AND DOCUMENTATION

This report documents the completed Tasks for Today Management System. The Codelgniter 4 project implements public task viewing, authenticated full CRUD operations, validation, session-based login, CSRF protection, and soft deletion. The screenshots provide direct evidence of the database, protected workflows, and the existing GitHub repository used for submission.

A. DATABASE IMPLEMENTATION

The project uses the tasks_today_tsa2 MySQL database. The migrations and SQL export create the users and tasks tables, while the seeder adds the demonstration administrator account and sample task data.

Table	Key fields	Purpose
users	id, username, password, full_name, email	Stores the login identity and one-way password hash.
tasks	id, title, description, status, task_date, is_archived	Stores task records and the soft-delete state.

Database Rules

Requirement	Implementation	Status
Password security	password_hash() output is stored and password_verify() checks login.	Passed
Task validation	Title and a valid task date are required before create or update.	Passed
Soft deletion	Archive sets is_archived to 1; records remain in MySQL.	Passed
Public filtering	Archived tasks are excluded from Welcome and Task List queries.	Passed

A. DATABASE EVIDENCE

The screenshot displays the phpMyAdmin web interface. The left sidebar shows a tree of databases including 'bizdash_db', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', 'pos_system', 'tasks_today', and 'test'. The main panel is titled 'Database: tasks_today_tsa2' and features a top navigation bar with tabs for Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, Triggers, and Designer. Below the navigation bar is a 'Filters' section with a search input. The central area shows a table list for the 'tasks_today_tsa2' database. It contains two tables: 'tasks' and 'users'. The 'tasks' table has 4 rows, InnoDB engine, utf8mb4_unicode_ci collation, and a size of 16.0 KiB. The 'users' table has 1 row, InnoDB engine, utf8mb4_unicode_ci collation, and a size of 32.0 KiB. Below the table list, there is a '2 tables Sum' section showing a total of 5 rows and a size of 48.0 KiB. At the bottom, there is a 'Create new table' section with input fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Table	Action	Rows	Type	Collation	Size	Overhead
tasks	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
users	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
2 tables Sum		5	InnoDB	utf8mb4_unicode_ci	48.0 KiB	0 B

Figure 1. tasks_today_tsa2 database containing users and tasks tables.

phpMyAdmin

Server: 127.0.0.1 > Database: tasks_today_tsa2 > Table: users

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(10)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	username	varchar(50)	utf8mb4_unicode_ci		No	None			Change Drop More
3	password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
4	full_name	varchar(100)	utf8mb4_unicode_ci		No	None			Change Drop More
5	email	varchar(150)	utf8mb4_unicode_ci		No	None			Change Drop More
6	created_at	datetime			Yes	NULL			Change Drop More
7	updated_at	datetime			Yes	NULL			Change Drop More

Check all With selected Browse Change Drop Primary Unique Index Spatial Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after updated_at Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	id	0	A	No	
Edit Rename Drop	users_username_unique	BTREE	Yes	No	username	0	A	No	

Create an index on 1 columns Go

Partitions

No partitioning defined

Partition table

Information

Space usage Row statistics

Console

Press Ctrl+Enter to execute query

Overhead	Next autoindex
Effective 32.0 KIB	Creation Oct 06, 2026 at 04:13 PM
Total 32.0 KIB	Last update Oct 06, 2026 at 10:49 AM
Optimize table	Last check Oct 06, 2026 at 10:49 AM

Figure 2. Users table structure including the password hash field.

A. DATABASE EVIDENCE

phpMyAdmin

Recent Favorites

Database: tasks_today_baz2 > Table: tasks

Table structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(10)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(150)	utf8mb4_unicode_ci		No	None			Change Drop More
3	description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
4	status	varchar(20)	utf8mb4_unicode_ci		No	pending			Change Drop More
5	task_date	date			No	None			Change Drop More
6	is_archived	tinyint(1)			No	0			Change Drop More
7	created_at	datetime			Yes	NULL			Change Drop More
8	updated_at	datetime			Yes	NULL			Change Drop More

Check all With selected Browse Change Drop Primary Unique Index Spatial Fulltext

Print Propose table structure Move columns Normalize

Add 1 column(s) after updated_at Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	id	3	A	No	

Create an index on 1 columns Go

Partitions

No partitioning defined!

Partition table

Information

Console Space usage Row statistics

Press Ctrl+Enter to execute query

Overhead	Effective	Total
0 B	16.0 KiB	16.0 KiB

Next autoindex	Creation	Last update	Last check
5	Oct 06, 2026 at 04:13 PM	Oct 06, 2026 at 10:49 AM	Oct 06, 2026 at 10:49 AM

Optimize table

Figure 3. Tasks table structure including task_date and is_archived.

phpMyAdmin

Server: 127.0.0.1 » Database: tasks_today_tsa2 » Table: tasks

Showing rows 0 - 3 (4 total, Query took 0.0002 seconds)

SELECT * FROM `tasks`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		id	title	description	status	task_date	is_archived	created_at	updated_at
☐	Edit Copy Delete	1	Review project requirements	Check all TSA2 requirements before implementation.	completed	2026-10-06	0	2026-10-06 16:13:23	2026-10-06 16:13:23
☐	Edit Copy Delete	2	Test authentication flow	Verify logged-out redirects and logged-in manageme...	pending	2026-10-06	0	2026-10-06 16:13:23	2026-10-06 16:13:23
☐	Edit Copy Delete	3	Prepare submission package	Review the repository, database, README, and docum...	pending	2026-10-07	0	2026-10-06 16:13:23	2026-10-06 16:13:23
☐	Edit Copy Delete	4	QA verification task updated	Created during the final end-to-end test.	completed	2026-10-08	1	2026-10-06 08:14:41	2026-10-06 08:14:48

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Figure 4. Task records retained in the database, including an archived QA record.

B. PROJECT STRUCTURE AND VERIFICATION

Component	Responsibility
app/Controllers/Login.php	Validates credentials, verifies the password hash, and manages the session.
app/Controllers/Tasks.php	Handles task list, create, edit, update, and archive workflows.
app/Filters/AuthFilter.php	Redirects logged-out users away from protected management routes.
app/Models/TaskModel.php	Defines allowed task fields and filters active records.
app/Config/Routes.php	Keeps reading routes public and applies auth to management routes.
app/Views	Provides the login, task list, form, profile, about, and dashboard pages.
database/tasks_today_tsa2.sql	Provides a ready-to-import database schema and demonstration data.

Functional Test Results

Test	Expected behavior	Result
Open /tasks/new while logged out	Redirect to /login	Passed
Submit invalid credentials	Show a login error without starting a session	Passed
Submit valid credentials	Regenerate session and open the authenticated task list	Passed
Create with missing required data	Show validation errors and retain submitted values	Passed
Create, edit, and update a valid task	Persist changes and show them in the active list	Passed
Archive a task	Retain row with is_archived = 1 and hide it publicly	Passed
Submit state-changing forms	Require valid CSRF tokens	Passed

C. SYSTEM PAGES AND TEST EVIDENCE

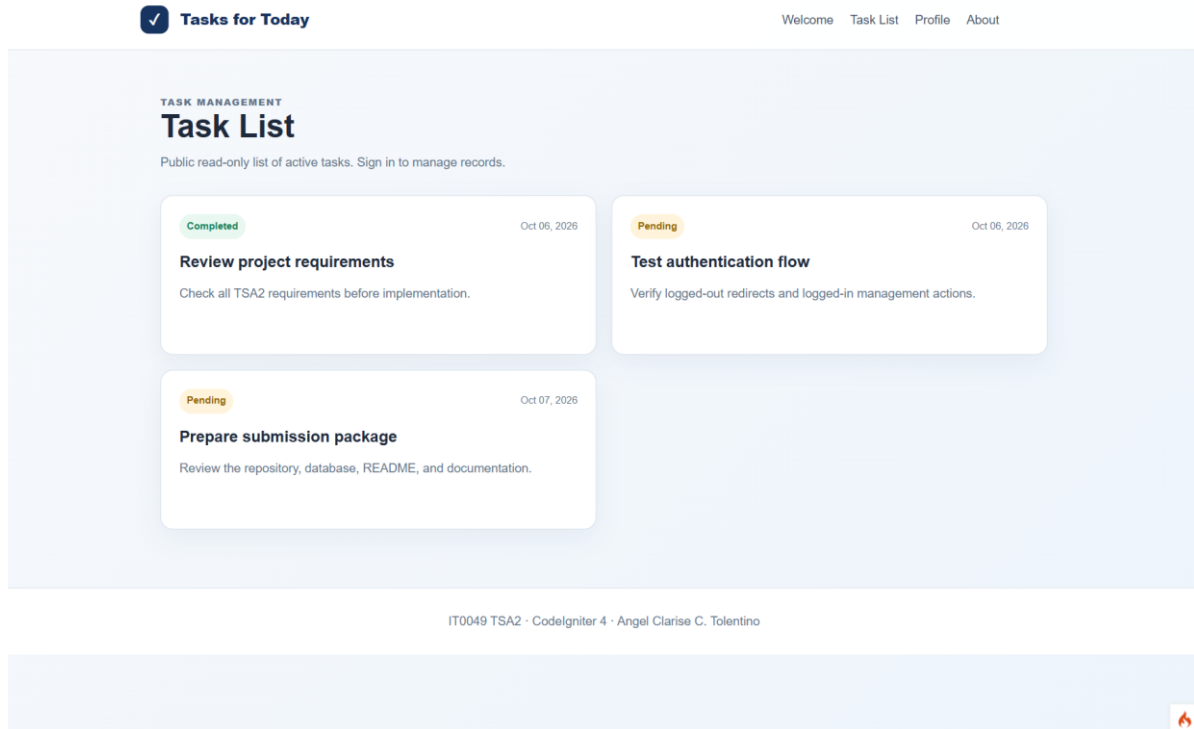


Figure 5. Public Task List shows active tasks without a login.

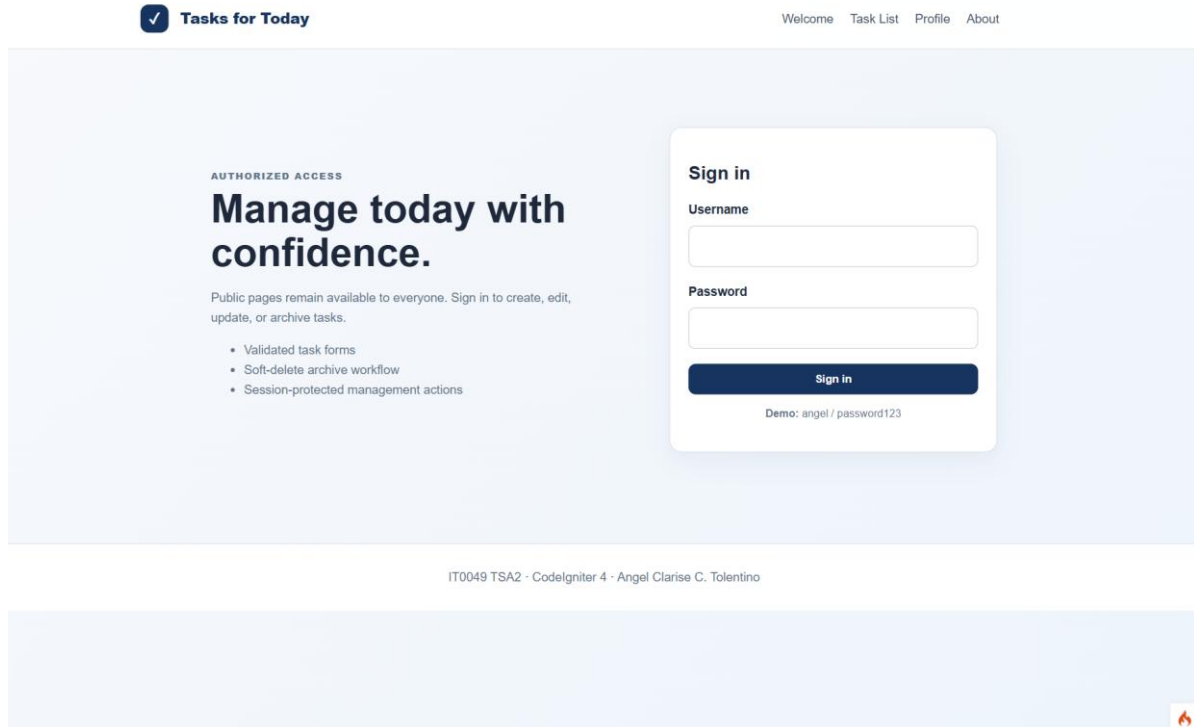


Figure 6. The protected new-task route redirects a logged-out user to Login.

C. SYSTEM PAGES AND TEST EVIDENCE

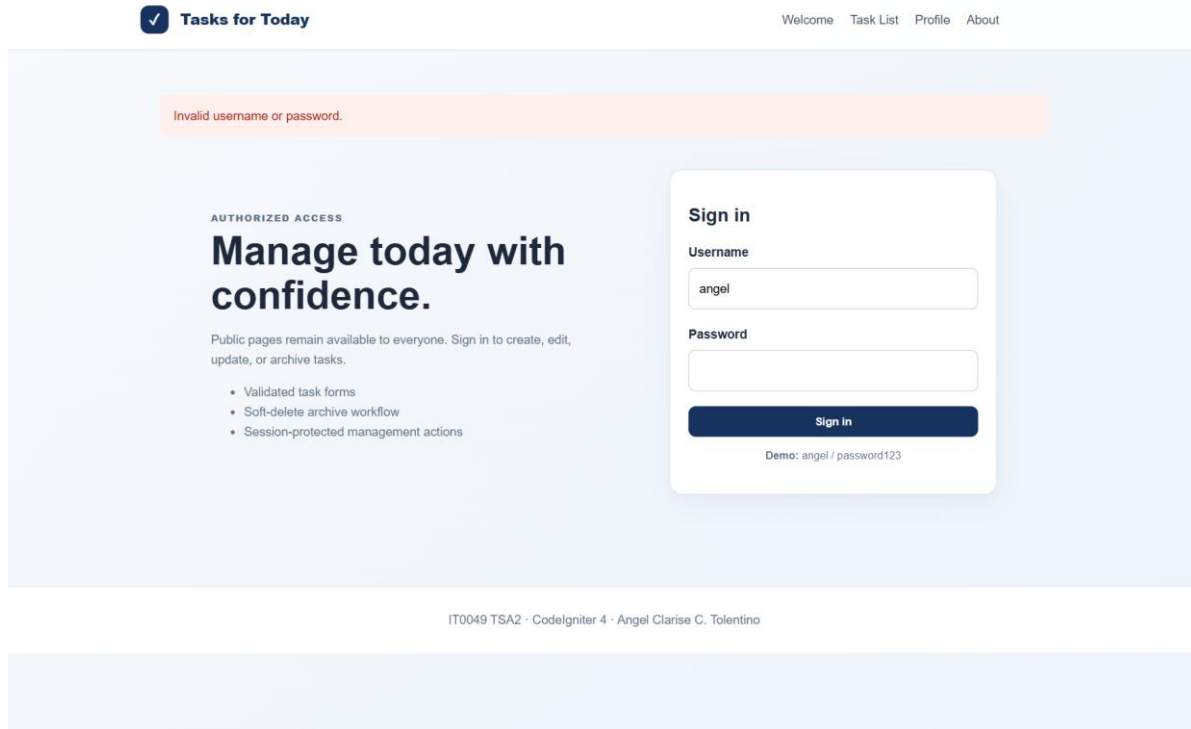


Figure 7. Invalid credentials display an authentication error.

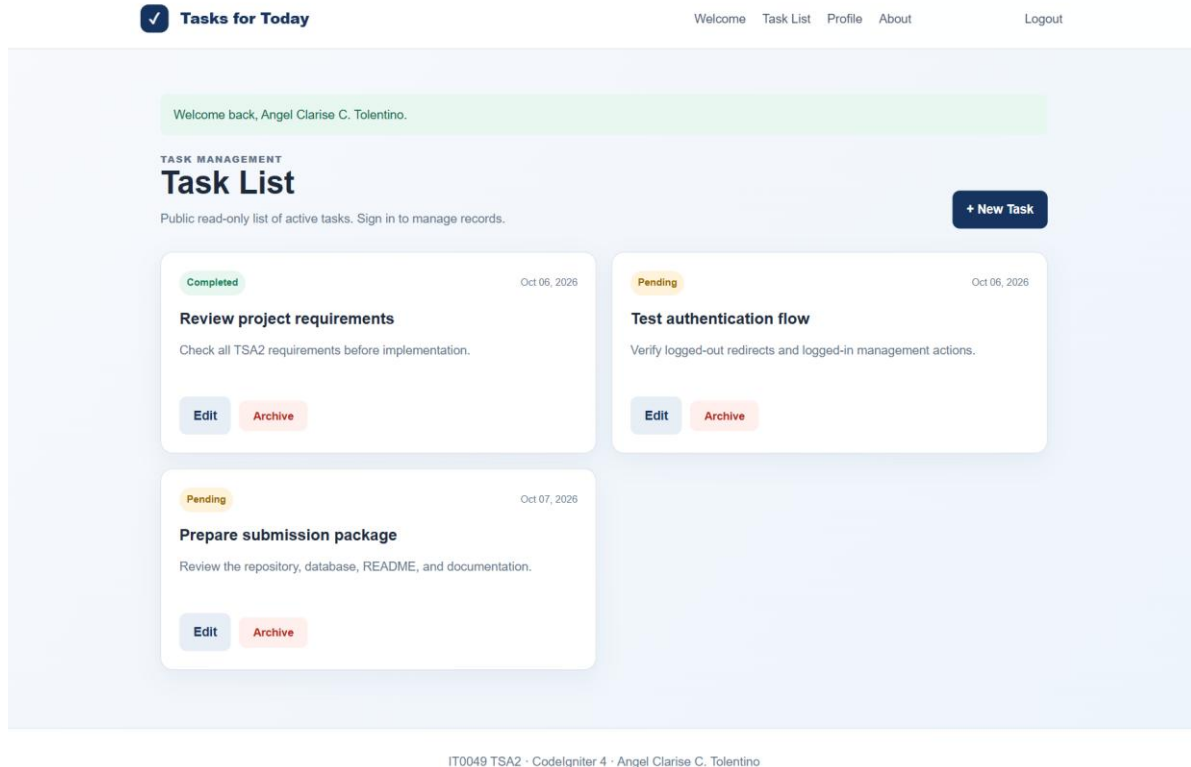


Figure 8. Authenticated task list with create, edit, and archive controls.

C. SYSTEM PAGES AND TEST EVIDENCE

The screenshot shows a web application interface for creating a task. At the top, there is a navigation bar with a logo 'Tasks for Today' and links for 'Welcome', 'Task List', 'Profile', 'About', and 'Logout'. The main content area has a header 'PROTECTED MANAGEMENT ACTION' and a title 'Create Task'. Below the title, a message states 'All required fields are validated before saving.' The form itself is a white box with a light blue border. It contains a 'Title' field with an asterisk, a 'Description' field, a 'Task date' field with a calendar icon and a placeholder 'dd/mm/yyyy', and a 'Status' dropdown menu with 'Pending' selected. At the bottom of the form are two buttons: 'Create Task' and 'Cancel'. The footer of the page includes the text 'IT0049 TSA2 · CodeIgniter 4 · Angel Clarise C. Tolentino' and a small red flame icon.

Figure 9. New Task form for authenticated users.

This screenshot shows the same 'Create Task' form as Figure 9, but with server-side validation applied. The 'Title' field now contains the text 'Validation screenshot task'. The 'Task date' field is empty, and a red error message 'The task_date field is required.' is displayed below it. The 'Status' dropdown remains set to 'Pending'. The 'Create Task' and 'Cancel' buttons are still present at the bottom of the form. The navigation bar and footer are identical to the previous screenshot.

Figure 10. Server-side validation of the required task date.

C. SYSTEM PAGES AND TEST EVIDENCE

Tasks for Today Welcome Task List Profile About Logout

PROTECTED MANAGEMENT ACTION

Edit Task

All required fields are validated before saving.

Title *

Review project requirements

Description

Check all TSA2 requirements before implementation.

Task date * 06/10/2026 **Status *** Completed

Save Changes **Cancel**

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Figure 11. Edit Task form populated with the selected record.

D. GITHUB SUBMISSION

Repository link: <https://github.com/angelc-byte/pos-system-tfa4>

The same course repository shown in the previous activity was continued, as instructed. No new repository was created. Commit 3329ec5 adds the completed TSA2 full CRUD and authentication implementation to the main branch.

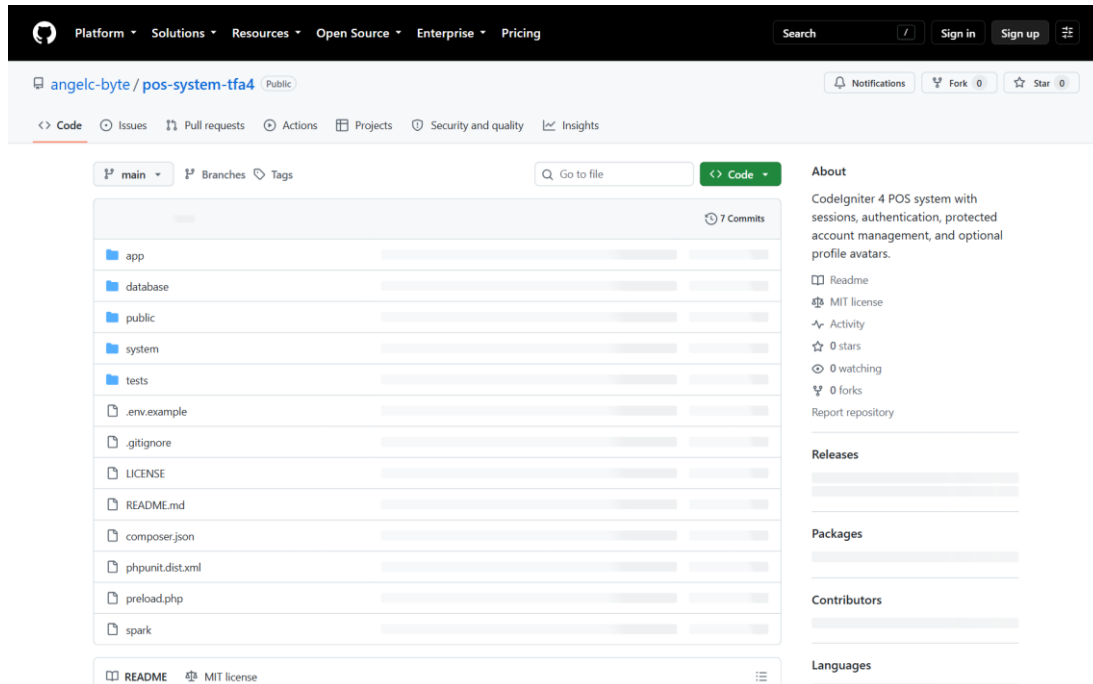


Figure 12. Existing course GitHub repository with the completed TSA2 commit.

Demonstration Credentials

Item	Value
Login URL	http://localhost/tasks-today-tsa2/public/login
Username	angel
Password	password123
Role	Administrator demonstration account

Local Setup

- Extract the project under C:\xampp\htdocs\tasks-today-tsa2.
- Copy .env.example to .env and confirm the MySQL settings.
- Import database/tasks_today_tsa2.sql in phpMyAdmin.
- Start Apache and MySQL, then open the local Login URL above.

Submission Package

The ZIP contains the CodeIgniter project, migrations, seeder, SQL export, README, and environment example. The documentation is supplied in editable DOCX and PDF formats. A hosted link is intentionally omitted because the instructor stated that deployment is not required.

CONCLUSION

The Tasks for Today Management System satisfies the required CodeIgniter authentication and CRUD outcomes. Public pages remain accessible, management actions require a verified session, task input is validated, archived rows are preserved and filtered, and the work is committed to the existing course repository. The included screenshots document the implemented behavior and database state needed for submission.